

Out No.:

Date:

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

TOPIC: Sensors and actuators

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Sensors and actuators

Actuator



Sensor



Introduction :

Modern electronic and embedded systems need a way to interact with the physical world. Computers and microcontrollers work with electrical signals and digital data, while the real world contains physical quantities such as temperature, pressure, light, humidity, motion, distance, sound, force and gas concentration. Sensors and actuators provide the connection between the physical world and an electronic system.

A sensor detects or measures a physical quantity and converts it into an electrical signal that can be understood and processed by an electronic system.

An actuator performs the opposite type of function. It receives an electrical or control signal from a controller and converts that signal into a physical action such as movement, rotation, switching, heating, cooling or pumping.

For example, consider an automatic room-temperature control system. A temperature sensor measures the room temperature and sends the information to a microcontroller. The microcontroller compares the measured temperature with the desired temperature. If the temperature is too high, it sends a signal to an actuator such as a motor or relay, which turns on a cooling fan.

Sensor :

A sensor is a device that detects a physical, chemical or biological quantity and converts it into a corresponding electrical signal.

The quantity being measured is called the measurand.

Examples of measurands include:

- Temperature
- Pressure
- Light intensity
- Humidity
- Distance
- Motion
- Sound
- Gas concentration
- Acceleration
- Force
- Magnetic field

For example, an **LM35 temperature sensor** detects temperature and produces an electrical voltage proportional to the temperature.

If the temperature increases, the output voltage also increases

1. Temperature Sensor

A **temperature sensor** is a device that detects the temperature of an object or environment and converts it into an electrical signal that can be measured by a controller.

Working

Temperature sensors work on the principle that the **electrical properties of a material change with temperature**.

For example:

- In an **RTD**, resistance increases as temperature increases.
- In a **thermistor**, resistance changes significantly with temperature.
- In an **LM35**, the output voltage changes proportionally with temperature.

Types

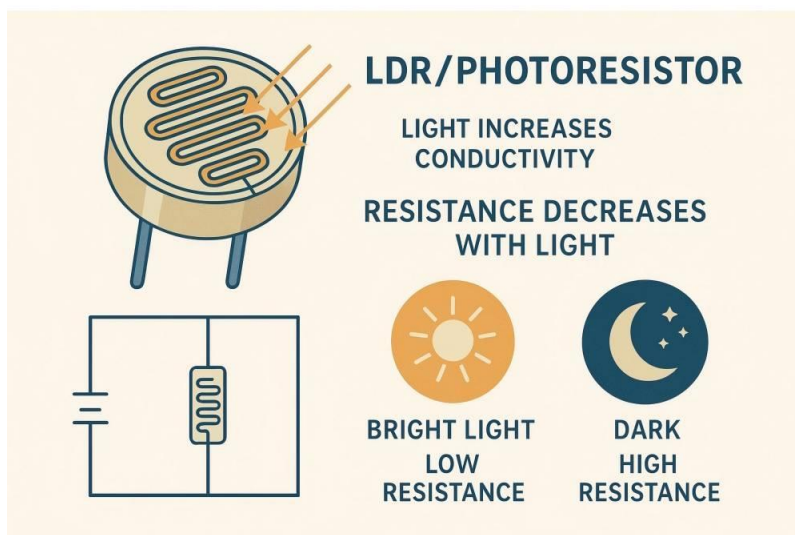
- **Thermistor**
- **RTD (Resistance Temperature Detector)**
- **Thermocouple**
- **LM35**
- **DS18B20**

Applications

- Air conditioners

- Refrigerators
- Industrial machines
- Weather monitoring
- IoT systems
- Fire detection systems

2. LDR (Light Dependent Resistor) An LDR is a sensor used to detect the intensity of light



Working

An LDR is made from a **photosensitive material**.

Its resistance changes according to the amount of light falling on it:

- **More light → resistance decreases**
- **Less light → resistance increases**

A microcontroller can measure the resulting voltage change and determine whether the environment is bright or dark.

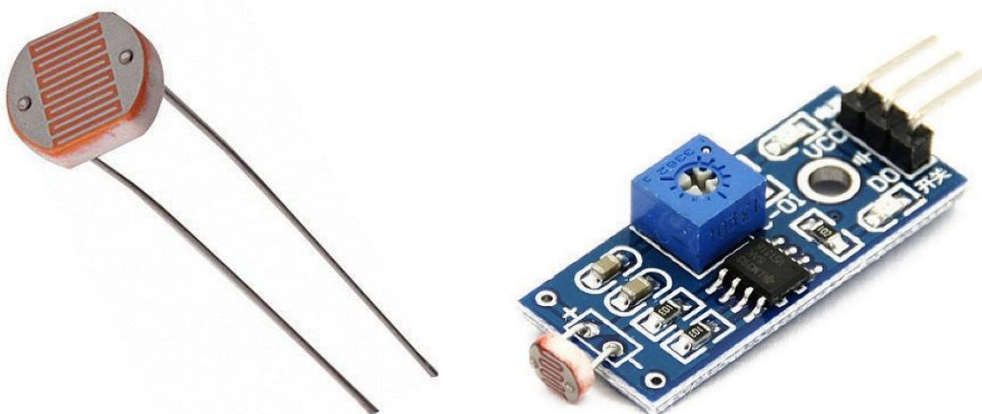
Applications

- Automatic street lights
- Automatic brightness control
- Security systems
- Light meters
- Smart home systems

Example

In an automatic street light:

- During daytime → LDR detects high light → light remains OFF.
- At night → LDR detects low light → light turns ON.



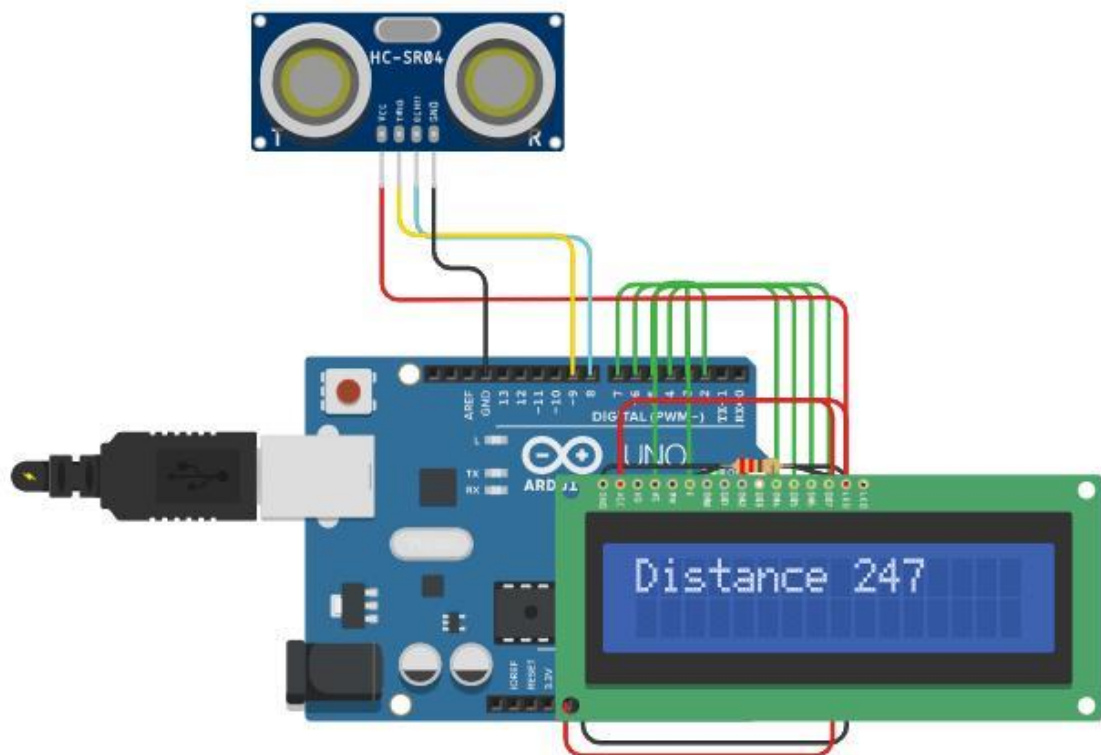
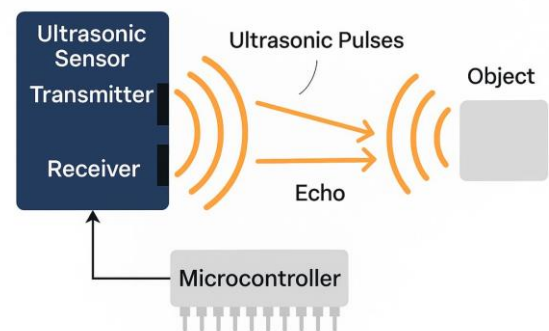
3. Ultrasonic Sensor

An **ultrasonic sensor** measures distance using high-frequency sound waves.

A popular example is the **HC-SR04**.



Ultrasonic Sensor Basic Working Principle



Working

The sensor sends an ultrasonic sound wave toward an object.

The sound wave:

1. Travels toward the object.
2. Hits the object.
3. Gets reflected back.
4. Is detected by the sensor.

The controller measures the **time taken for the echo to return**.

The distance can be calculated as:

$$\text{Distance} = (\text{Speed of sound} \times \text{Time}) / 2$$

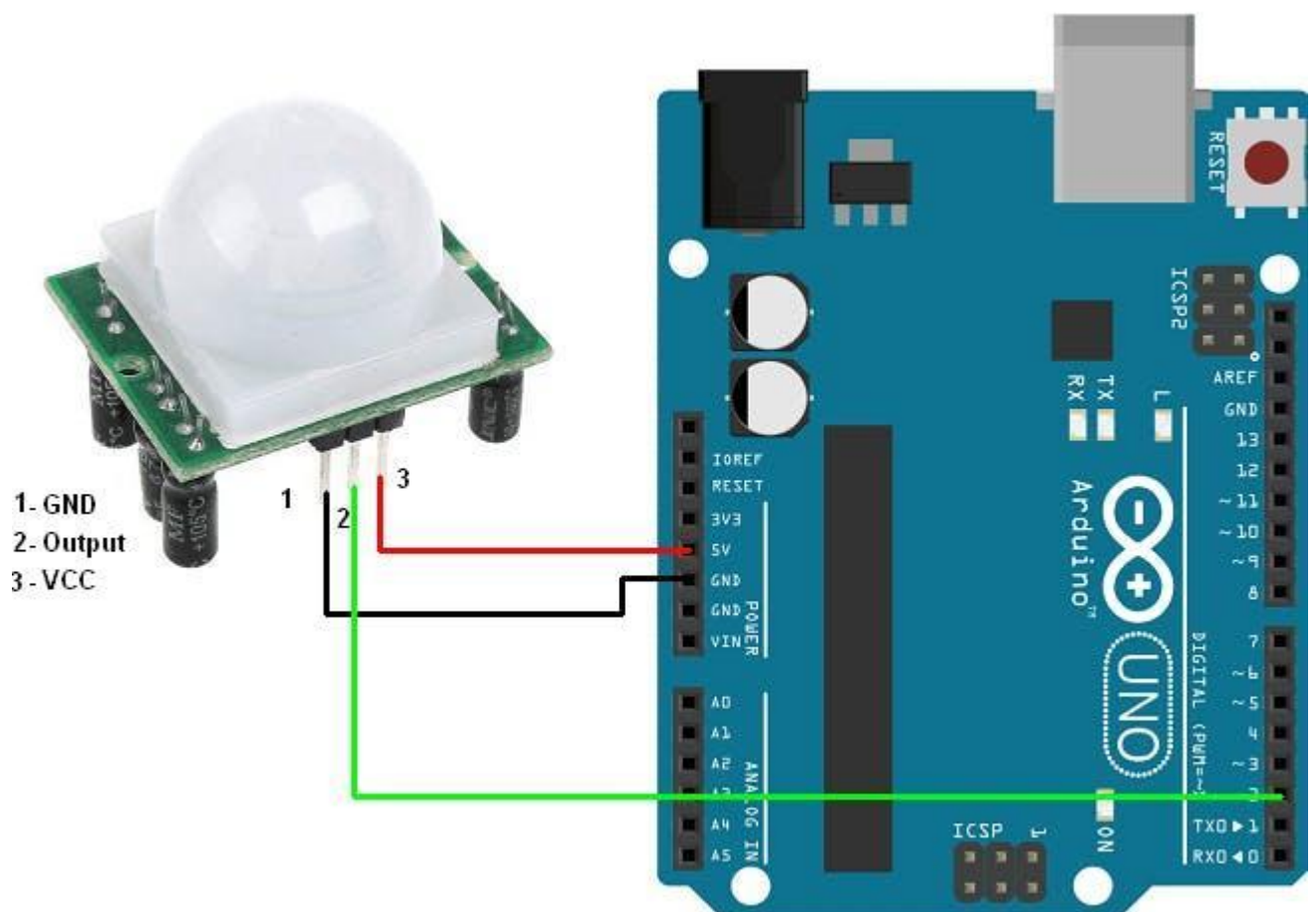
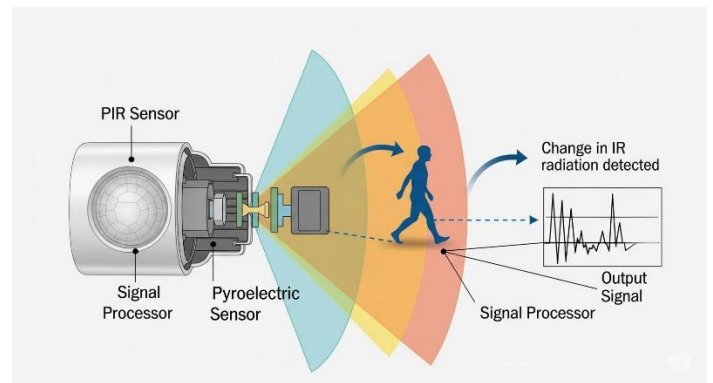
We divide by 2 because the sound travels **to the object and back**.

Applications

- Obstacle detection
- Parking sensors
- Robot navigation
- Water-level measurement
- Automatic doors
- Smart dustbins

4. PIR Motion Sensor

A **PIR (Passive Infrared)** sensor detects movement of humans or animals by sensing changes in infrared radiation.



Working

Human bodies emit infrared radiation because of their body temperature.

A PIR sensor contains a **pyroelectric element** that detects changes in infrared radiation.

When a person moves in front of the sensor:

Human movement → change in IR radiation → sensor output changes → motion detected

Applications

- Security alarms
- Automatic lights
- Smart homes
- Intruder detection
- Automatic doors

Important point

A PIR sensor generally detects **movement**, rather than simply determining whether a person is present.

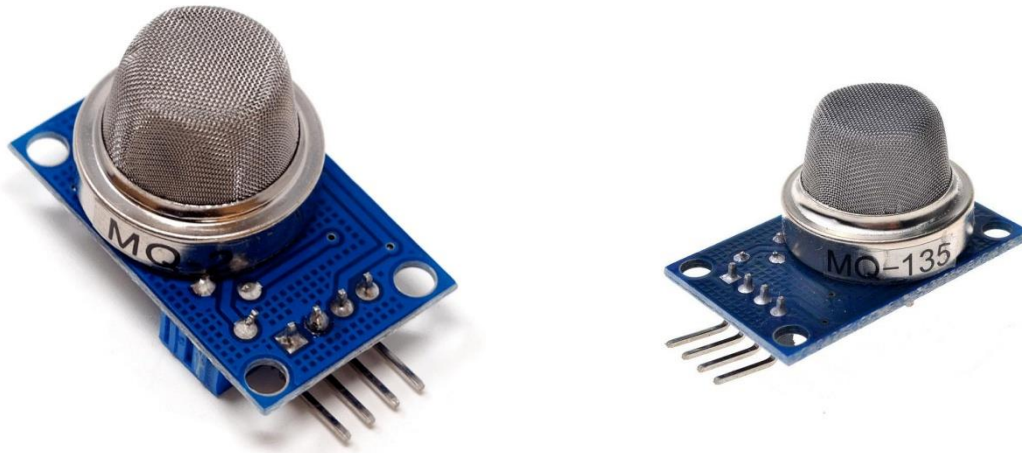
5. Gas Sensor

Gas sensors detect the presence or concentration of particular gases in the environment.

Examples include:

- **MQ-2** → smoke, LPG, methane

- **MQ-3** → alcohol
- **MQ-7** → carbon monoxide
- **MQ-135** → air-quality-related gases



Working

Many MQ sensors contain a sensing material whose **electrical resistance changes when exposed to certain gases.**

The sensor produces an electrical output corresponding to the gas exposure.

Applications

- Gas leakage detection
- Air-quality monitoring
- Industrial safety
- Pollution monitoring
- Smart homes
- Fire and disaster monitoring

Actuators

An **actuator** is a device that converts an **electrical, pneumatic, or hydraulic signal into physical movement or mechanical action.**

In simple words:

Sensor → detects something

Controller → makes a decision

Actuator → performs the action

Example

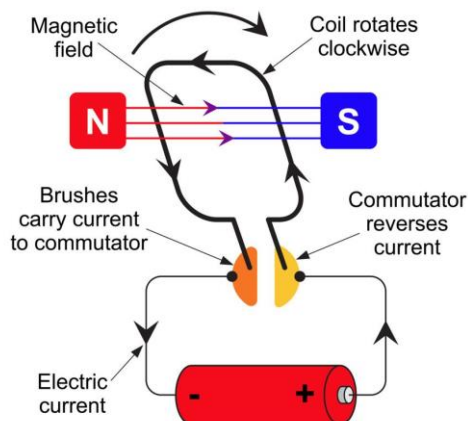
In an automatic door:

**Motion sensor → detects person → Controller → Actuator
→ Door opens**

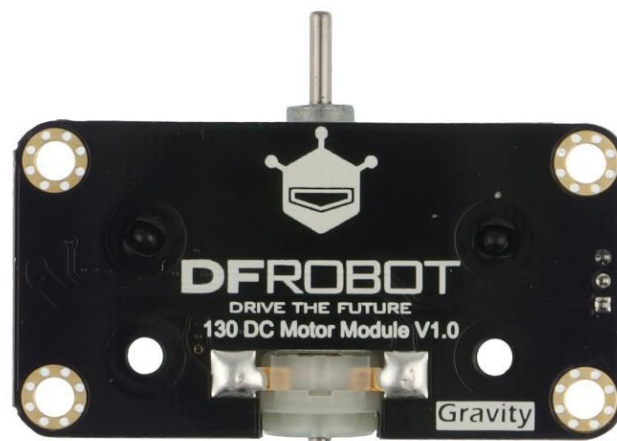
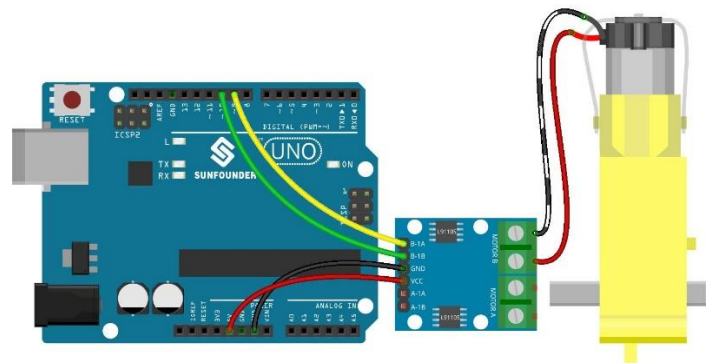
So, actuators are basically the “**action-performing**” **components** of an automation or IoT system.

1. DC Motor

A **DC motor** is an actuator that converts electrical energy into **rotational mechanical energy**.



A basic setup of a DC motor.



Working

When DC voltage is supplied to the motor, current flows through its windings. The interaction between the **magnetic field** and current-carrying conductors produces a force that causes the motor shaft to rotate.

The direction of rotation can be changed by reversing the polarity of the supply.

Applications

- Robots
- Electric fans
- Conveyor belts
- Drones
- Toys
- Pumps
- Automated machines

Advantages

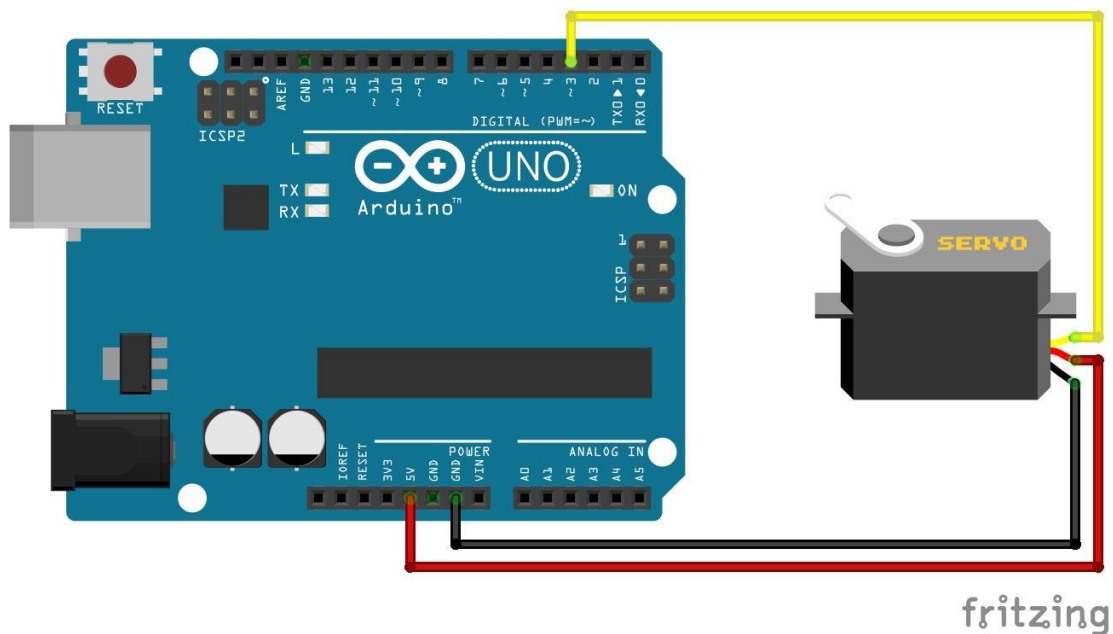
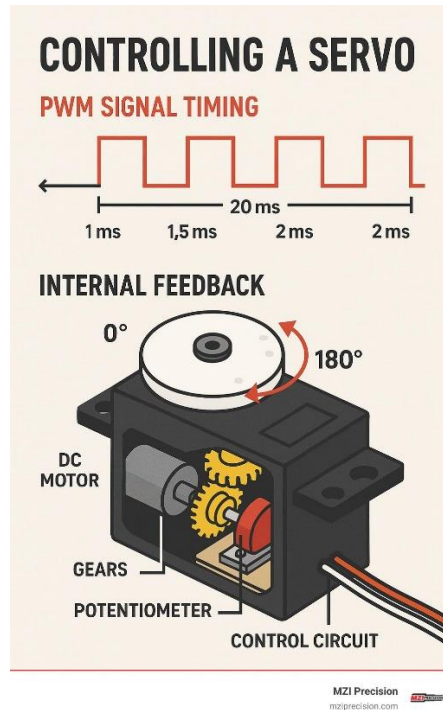
- Simple construction
- Easy to control
- High starting torque
- Relatively inexpensive

Disadvantages

- Brushes may wear out
- Requires maintenance
- Produces electrical noise

2. Servo Motor

A **servo motor** is an actuator used when **precise angular positioning** is required.



Working

A servo motor usually consists of:

- DC motor

- Gear mechanism
- Position sensor
- Control circuit

The controller gives the servo a control signal. The internal circuit compares the **desired position** with the **actual position** and rotates the motor until the required position is reached.

For example:

Controller → 90° command → Servo → rotates to approximately 90°

Applications

- Robotic arms
- Camera gimbals
- RC cars
- Drones
- Automatic doors
- Robotic projects

Advantages

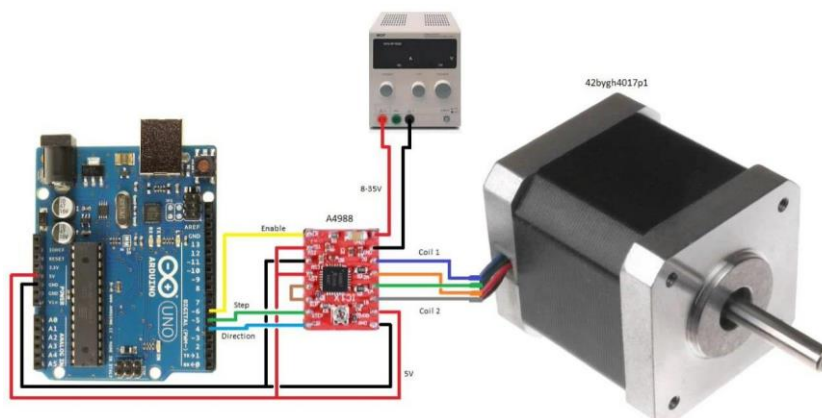
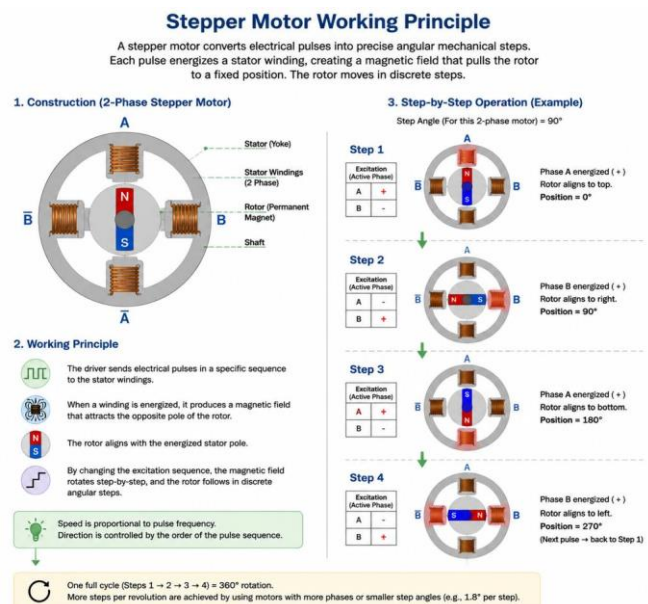
- Precise positioning
- Good control
- Compact
- Easy to interface with microcontrollers

Disadvantages

- Limited rotation in many hobby servos
- More expensive than basic DC motors
- Cannot always provide continuous rotation

3. Stepper Motor

A **stepper motor** is an actuator that rotates in **small, fixed steps** rather than rotating continuously like a conventional DC motor.



Working

The motor contains multiple electromagnetic coils.

The controller energizes these coils in a particular sequence.

For example:

Coil A → Coil B → Coil C → Coil D

This causes the rotor to move step by step.

Because the controller knows how many steps were given, it can control the approximate position of the motor.

Applications

- 3D printers
- CNC machines
- Printers
- Robotics
- Camera systems
- Automated positioning systems

Advantages

- Precise movement
- Good positioning
- Easy digital control
- Can hold a position when energized

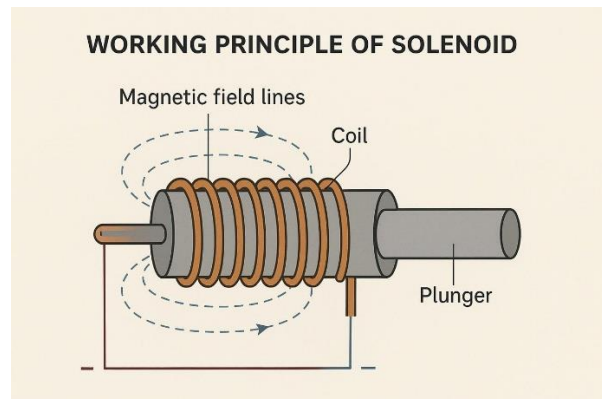
Disadvantages

- Can lose steps if overloaded
- Consumes power even while holding position

- Can produce vibration and noise

4. Solenoid

A **solenoid** is an electromagnetic actuator that converts electrical energy into **linear motion**.



Working

A solenoid consists mainly of:

- Coil of wire
- Iron/plunger
- Housing

When current flows through the coil, it creates a magnetic field.

The magnetic field pulls the iron plunger into the coil.

So:

Electrical current → Magnetic field → Plunger movement

When the current is removed, the plunger generally returns using a spring or another mechanical force.

Applications

- Electronic locks
- Door locks
- Valves
- Automatic machines
- Car systems
- Vending machines

Example

In an electronic door lock:

Correct password → Controller → Solenoid activated → Lock mechanism moves → Door unlocks

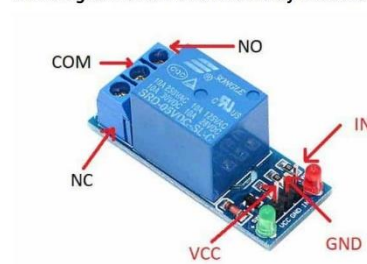
5. Relay

A **relay** is an electrically operated switching actuator.

It allows a low-power control circuit to switch another electrical circuit.



Pin diagram of a 1-Channel relay module



Working

A traditional electromagnetic relay contains:

- Coil
- Electromagnet
- Armature
- Contacts

When current flows through the coil, it creates a magnetic field.

The magnetic field moves the armature, changing the electrical contacts.

A relay commonly has:

- **COM** – Common
- **NO** – Normally Open
- **NC** – Normally Closed

Applications

- Switching lights
- Switching motors
- Home automation
- Industrial control
- Electrical protection/control systems

Example

An IoT system detects that the room is dark:

LDR → Microcontroller → Relay → Light ON

The relay itself doesn't normally provide the power to the lamp; it **controls the electrical connection** to it.

6. Pneumatic Actuator

A **pneumatic actuator** uses **compressed air** to produce mechanical movement.



Working

Compressed air is supplied to a cylinder.

The pressure acts on a piston and produces linear movement.

Compressed air → Pressure → Piston movement → Mechanical action

There are two common types:

Single-acting cylinder

Air pressure moves the piston in one direction, while a spring returns it.

Double-acting cylinder

Compressed air is supplied alternately to both sides of the piston, allowing movement in both directions.

Applications

- Industrial automation
- Packaging machines
- Robotic systems
- Manufacturing machines
- Clamping systems

Advantages

- Fast operation
- Clean operation
- Suitable for repetitive industrial work
- High force-to-weight ratio

7. Hydraulic Actuator

A **hydraulic actuator** uses **pressurized liquid**, usually hydraulic oil, to generate mechanical movement.



Working

A hydraulic pump pressurizes the hydraulic fluid.

The pressurized fluid enters a cylinder and pushes a piston.

Pump → Pressurized fluid → Hydraulic cylinder → Piston movement

Hydraulic systems can generate very large forces.

Applications

- Excavators
- Cranes
- Construction equipment
- Hydraulic presses
- Heavy machinery
- Aircraft systems

Advantages

- Very high force
- Smooth movement
- Suitable for heavy loads

Disadvantages

- Oil leakage can occur
- More complex system
- Requires maintenance
- Heavier than many electrical systems

8. Linear Actuator

A **linear actuator** produces movement in a **straight line**.

Instead of rotating a shaft continuously, it pushes or pulls an object.



Working

An electric motor rotates a screw mechanism.

The screw converts the **rotary motion of the motor into linear motion**.

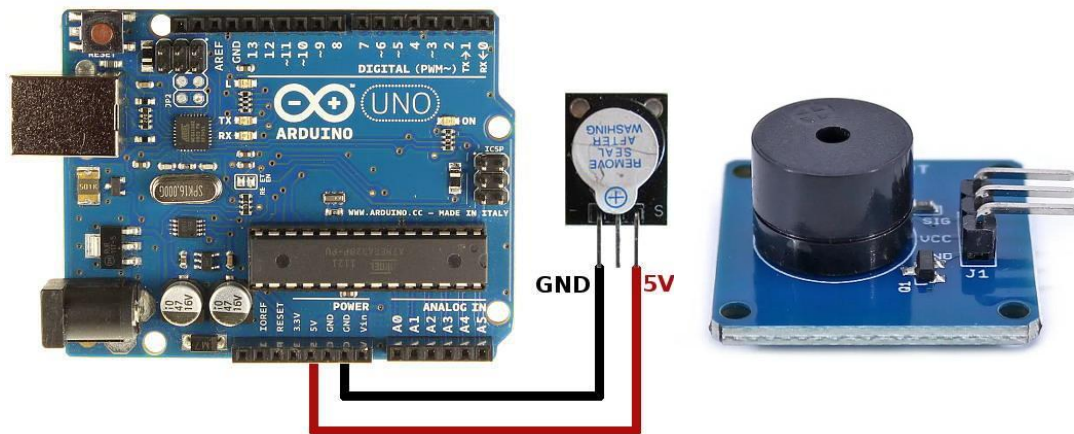
Motor rotation → Screw rotation → Nut moves → Linear movement

Applications

- Adjustable beds
- Automatic windows
- Robotic systems
- Solar tracking systems
- Electric lifting systems
- Industrial machines

9. Buzzer

A **buzzer** is an actuator that converts electrical energy into sound.



Working

When an electrical signal is applied, the internal component vibrates and produces sound.

Applications

- Alarm systems
- Fire alarms
- Security systems
- Electronic devices
- IoT warning systems

Example

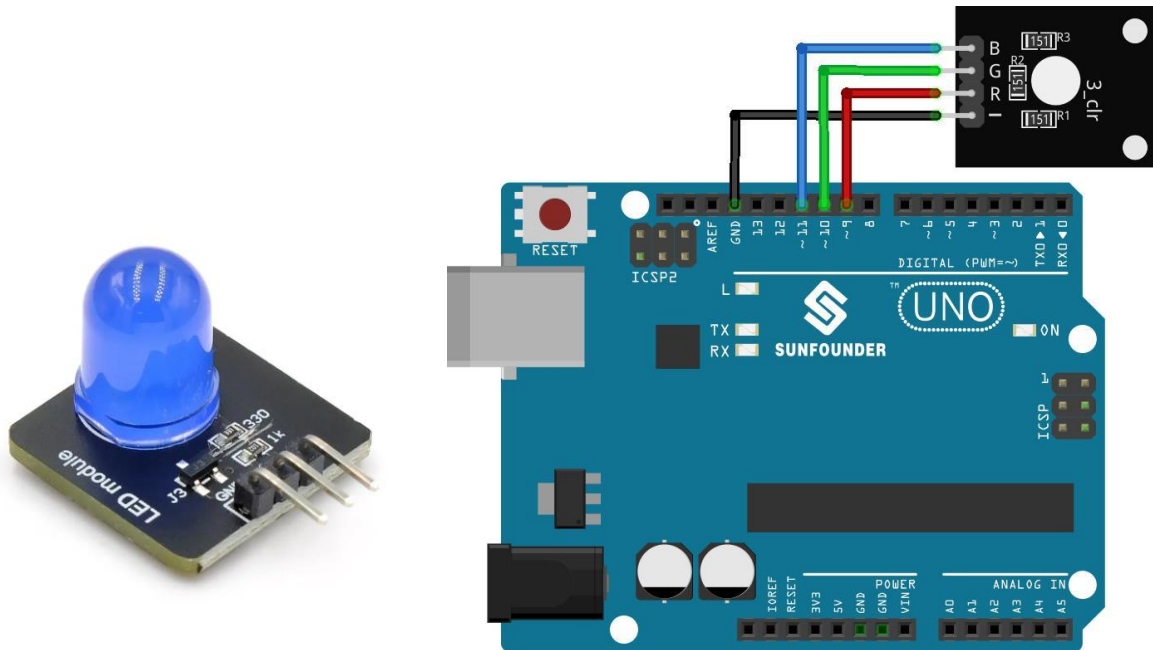
A gas sensor detects leakage:

Gas sensor → Controller → Buzzer → Alarm sound

10. LED

An **LED (Light Emitting Diode)** can also act as an actuator when it is used to provide a visual output.

It converts electrical energy into **light**.



Applications

- Status indicators
- Warning systems
- Displays
- Smart lighting
- IoT devices

For example:

Temperature sensor → Controller → LED → Warning indication

Sensor vs Actuator

Feature	Sensor	Actuator
Main function	Detects/measures	Performs an action
Converts	Physical quantity → electrical signal	Electrical/control signal → physical action
Direction	Input	Output
Example	Temperature sensor	Motor
Other examples	LDR, PIR, ultrasonic	Relay, servo, solenoid
Used for	Monitoring	Controlling

Simple IoT System

Physical Environment → Sensor → Microcontroller → Actuator → Physical Action

For example, in a **smart fire-monitoring system**:

Temperature sensor + Smoke sensor → ESP32 → Relay/Buzzer/Fan

Here:

- Temperature sensor = **input**
- Smoke sensor = **input**
- ESP32 = **controller**
- Relay = **actuator**

- Fan = **device being controlled**
- Buzzer = **output actuator**